Polluted Black Hole Companions: Spallogenic Lithium from Hot Accretion Flows with 3-D Weighted Monte Carlo Models

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Metal pollution in white dwarf photospheres records the composition of accreted planetary material. In the polluted white dwarf GALEX J2339−0424, Doyle et al. (2021) attributed an excess of beryllium to the spallation of icy material by MeV protons in the radiation belt of a giant planet. We investigate whether an analogous process operates in black hole binaries. In quiescence, the inner accretion flow onto a stellar-mass black hole is expected to be a two-temperature, advection-dominated flow in which the ion temperature reaches 4–20 MeV within 20 Schwarzschild radii, sufficient for $\alpha+\alpha$ fusion and for proton spallation of CNO nuclei into lithium and beryllium. We extend the weighted Monte Carlo code SERPENS with an absorbing boundary at the innermost stable circular orbit and follow ions thermally evaporated from the flow, comparing the first post-Newtonian (1PN) treatment of REBOUNDx with exact Schwarzschild geodesics. Approximately 1.5% of the evaporated ions escape. Exact and Newtonian dynamics yield the same fraction, whereas the 1PN approximation overestimates it by a factor of 3.7 at $10\,r_s$. Combined with the Roche-lobe geometry of the binaries, the fraction of accreted gas delivered to the companion is $(0.9\text{–}1.5)\times10^{-4}$ for A0620−00, XTE J1118+480 and V404 Cyg, comparable to the upper end of the range required by Yi & Narayan (1997), who treated this quantity as a free parameter. Using accretion rates from published spectral fits and the observed outburst histories, the predicted surface lithium abundances are consistent with the measurements for A0620−00 and V404 Cyg and with the upper limit for XTE J1118+480, for depletion times of 4×10^7–3×10^8 yr. For the dormant Gaia black holes the delivered lithium is negligible, in agreement with the normal abundances of their companions. The model further predicts a beryllium abundance approximately five times the solar value in the companion of V404 Cyg, which can be tested with near-ultraviolet spectroscopy.

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I. INTRODUCTION

Metal-polluted white dwarfs provide a direct record of accreted planetary material. Because heavy elements sink out of white dwarf photospheres on timescales of days to millions of years, any metals detected in the photosphere must have been accreted recently. Doyle, Desch & Young [1] showed that this record can also carry a nuclear signature. The polluted white dwarf GALEX J2339−0424 exhibits a beryllium abundance, relative to iron, roughly 200–500 times the CI-chondritic value. Since beryllium is produced primarily by spallation, in which energetic protons fragment heavier nuclei such as oxygen, Doyle et al. attributed the excess to the irradiation of icy material. The inferred number ratio $n_{\rm Be}/n_{\rm O}\sim10^{-5}$ requires a proton fluence of $\sim10^{21}$ cm^{-2}, which ice accumulates in $\sim10^7$ yr within a Jovian radiation belt. The beryllium therefore traces the environment in which the accreted body resided.

SERPENS was developed to model trace material lost from volcanically active exo-Io satellites into the environments of close-in giant planets [2, 3]. In this work we apply the reasoning of Doyle et al. to black holes. Although a black hole cannot retain a chemical record, its binary companion can, and the inner accretion flow of a quiescent black hole constitutes a considerably more intense radiation environment than a planetary magnetosphere. At accretion rates well below the Eddington rate, the inner flow is expected to be advection dominated, with ions near the virial temperature [4–6]. Within $20\,r_s$ the ion temperature reaches 4–20 MeV, and the high-energy tail of the distribution exceeds the thresholds for $\alpha+\alpha\rightarrow{}^{6,7}$Li and for proton spallation of carbon, nitrogen and oxygen.

Lithium enhancements are indeed observed in the companions of several black hole X-ray transients. Martín et al. [7, 8] detected strong Li I 6708 Å absorption in V404 Cyg, A0620−00, Nova Mus 1991 and the neutron star system Cen X-4, and lithium has since been detected in four black hole companions [9]. The K-type secondary of A0620−00, for example, has $A(\mathrm{Li})=2.31\pm0.21$, compared with ≤0.35 for field stars of similar mass [10]; in stars this cool, convective mixing during the pre-main sequence and main sequence is expected to deplete the surface lithium [10]. Yi & Narayan [11] proposed that the lithium is synthesised by α–α reactions in the hot flow and subsequently transported to the companion. Their estimate required that 10^{-6}–10^{-4} of the accreted mass reach the secondary after processing; lacking a means of computing this fraction, they treated it as a free parameter. They further noted that accretion flows onto white dwarfs do not reach the required temperatures, so that such lithium is specific to neutron star and black hole systems.

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The delivered fraction is fundamentally a test-particle problem, which SERPENS is designed to address. We launch ions from the hot flow using the SERPENS thermal evaporation mode, follow them through the strong-field region with an absorbing boundary at the innermost stable circular orbit (ISCO), compute the proton and α fluences they accumulate, and determine the fraction of processed gas intercepted by the companion. We compare the resulting surface lithium abundances with the measurements in A0620−00, XTE J1118+480 and V404 Cyg, and with the companions of the three dormant Gaia black holes, which have been studied spectroscopically in detail. The G-dwarf companion of Gaia BH1 has $A(\mathrm{Li}) = 2.3 \pm 0.1$, which El-Badry et al. [12] consider typical of a solar-type star older than the Hyades, and an abundance pattern characteristic of the thin disk. The red-giant companion of Gaia BH2 is α-enhanced, $[\alpha/\mathrm{Fe}] = +0.26 \pm 0.05$, but otherwise normal and shows no lithium [13], and the metal-poor giant in Gaia BH3 shows no chemical peculiarities [14, 15].

The geometry of the problem is illustrated in Figure 1. Section II summarises the relevant spallation physics and relativistic length scales; Section III describes the extension of SERPENS, the hot-flow model and the delivery calculation; Section IV presents our results and their comparison with observations; Section V discusses the uncertainties; and Section VI summarises our conclusions. A benchmark of the integrator at binary scales is given in Appendix A.

II. SPALLATION NEAR A BLACK HOLE

A. The polluted white dwarf argument

For target nuclei exposed to a fluence F of projectiles above the reaction threshold, the number of product nuclei per target nucleus is, to first order,

$$\frac{n_{\mathrm{Be}}}{n_{\mathrm{O}}} = \sigma_{\mathrm{Be}} F, \qquad F = \int \Phi_{>}\, dt, \tag{1}$$

provided that $\sigma F \ll 1$. Here $\Phi_{>}$ denotes the flux of protons above ~ 10 MeV and $\sigma_{\mathrm{Be}} \approx 10^{-26}$ cm^2 is the cross section for $^{16}\mathrm{O}(p,x)^{9}\mathrm{Be}$ [1]. Equation (1) is the relation adopted by Doyle et al. for GALEX J2339−0424; we apply it without modification, the only difference being the source of the energetic protons.

B. The hot accretion flow as a radiation belt

In a two-temperature advection-dominated accretion flow (ADAF), the ions are heated by viscous dissipation and transfer energy to the electrons only through inefficient Coulomb collisions; as a result, they remain close to the virial temperature [4, 5]. We adopt

$$kT_i(r) = \theta_i \frac{m_p c^2}{x}, \qquad x \equiv r/r_s, \tag{2}$$

with $\theta_i = 0.55/6$, corresponding to 55% of the virial value $m_p c^2/(6x)$. This yields $kT_i \approx 86\,\mathrm{MeV}/x$, or 8.6 MeV at $10\,r_s$. The radial inflow velocity is $v_r = c_1\,\alpha\, v_K$, where $v_K = (GM/r)^{1/2} = c/\sqrt{2x}$, and we adopt $\alpha = 0.1$ and $c_1 = 0.5$ [5]. Mass conservation then determines the ion density,

$$n(r) = \frac{\dot{M}}{4\pi r^2\, v_r\, m_p}, \qquad \dot{M} = \dot{m}\dot{M}_{\mathrm{Edd}}, \tag{3}$$

where $\dot{M}_{\mathrm{Edd}} = L_{\mathrm{Edd}}/(0.1\,c^2)$ and $L_{\mathrm{Edd}} = 4\pi GMm_p c/\sigma_T$. The flux of ions with energies above a threshold E_{th} in a Maxwellian distribution at temperature T_i is

$$\Phi_{>}(r) = n_j(r)\, \frac{\Gamma(3/2, E_{\mathrm{th}}/kT_i)}{\Gamma(3/2)}\, \bar{v}_{>}, \tag{4}$$

where n_j is the number density of species j ($n_\alpha = 0.085\,n$), $\Gamma(a,y)$ is the upper incomplete gamma function, and $\bar{v}_{>}$ is the mean speed of ions above the threshold. We adopt $E_{\mathrm{th}} = 10$ MeV for proton spallation of oxygen and $E_{\mathrm{th}} = 35$ MeV (laboratory frame) for $\alpha + \alpha \rightarrow$ Li.

Since $n \propto \dot{m}/M$ and all dynamical timescales scale with M, the fluence accumulated by a fluid element is independent of the black hole mass and proportional to $\dot{m}$. The irradiation problem is therefore characterised by a single parameter, the Eddington-scaled accretion rate.

C. Relativistic length scales

For a non-spinning black hole, two radii delimit the region of interest. The event horizon lies at $r_s = 2GM/c^2$ (29.5 km for $10\,M_\odot$), and the ISCO lies at $r_{\mathrm{ISCO}} = 3\,r_s$ [16]. An ion that crosses the ISCO on an inward trajectory cannot return on a stable orbit; SERPENS therefore removes such particles and records them as accreted. In our model, spallation occurs predominantly between 3 and $\sim 30\,r_s$, where the escape speed is 0.2–0.6c. In this regime the adequacy of the gravitational treatment cannot be assumed, and we test it explicitly in Section IV A.

D. Applicability of a test-particle treatment

SERPENS is a collisionless code. A fluid description is appropriate when the Knudsen number $\mathrm{Kn} = \lambda_{\mathrm{mfp}}/L \ll 1$, whereas a kinetic treatment is required for $\mathrm{Kn} \gtrsim 1$ [17]. ADAFs fall in the latter regime: the Coulomb collision time exceeds the inflow time, which is why the ions and electrons maintain different temperatures [4, 6]. Our approach does, however, neglect the magnetic field, which couples the ions to the flow on gyration scales far smaller than r_s. We therefore treat the ballistic escape of each ion as a single evaporation attempt and interpret the resulting escape fraction as an estimate of the processed material available to an outflow, rather than as a solution of the magnetohydrodynamic problem. We return to this point in Section V.

III. METHODS

A. SERPENS

SERPENS [3] is a three-dimensional weighted Monte Carlo code that follows superparticles launched from a source body. Each superparticle is assigned a species, a launch velocity drawn from a physically motivated distribution, and a weight determined by the mass-loss rate of the source. Trajectories are integrated with REBOUND [18], and particles that collide with a massive body are removed. In the thermal evaporation mode, launch speeds follow a Maxwell–Boltzmann distribution,

$$f_{\rm th}(v) = \sqrt{\frac{2}{\pi}}\,\frac{v^2}{v_{\rm sc}^3}\exp\left(-\frac{v^2}{2v_{\rm sc}^2}\right), \qquad v_{\rm sc} = \sqrt{\frac{k_B T}{m}}. \tag{5}$$

B. Launching ions from the hot flow

We take a sphere of radius r_0 centred on the black hole as the source surface. Launch positions are drawn uniformly over this sphere using the SERPENS position sampler, and each ion is launched radially with a speed drawn from equation (5) at $T = T_i(r_0)$ and $m = m_p$. The rotation of the flow, $\Omega = 0.4\,\Omega_K$ about the orbital axis [5], is added to the launch velocity. We consider launch radii between 4 and $50\,r_s$. Because most of the lithium is produced within $20\,r_s$, we quote results for $r_0 = 10\,r_s$ and adopt 5–$20\,r_s$ as the range of uncertainty. Lengths are expressed in units of r_s and times in units of r_s/c, so that the dynamics are independent of the black hole mass.

C. Gravity near the black hole

Trajectories are integrated with the adaptive IAS15 integrator [19]. Because the REBOUND collision search scales as N^2 even for test particles, the ions are integrated in parallel batches of 150–400. The black hole is represented as a massive body with a collision radius equal to $r_{\rm ISCO}$, and the `line` collision search identifies ions that cross the ISCO between integration steps. Each launch radius is treated with three gravitational models:

1. Newtonian gravity in SERPENS;
2. SERPENS with the REBOUNDx `gr` force [20], which implements the single-source 1PN acceleration of Anderson et al. [21],

$$\mathbf{a} = -\frac{GM}{r^2}\hat{\mathbf{r}} + \frac{GM}{c^2 r^2}\left[\left(\frac{4GM}{r} - v^2\right)\hat{\mathbf{r}} + 4\,(\hat{\mathbf{r}}\cdot\mathbf{v})\,\mathbf{v}\right]; \tag{6}$$

3. exact Schwarzschild geodesics, evaluated analytically.

For an isolated black hole, the third case does not require numerical integration. Taking the launch speed to be that measured by a static observer, each ion has a conserved specific energy and angular momentum (in units with $c = r_s = 1$),

$$\mathscr{E} = \gamma\sqrt{1 - 1/x}, \qquad \mathscr{L} = \gamma x\, v_t, \tag{7}$$

and moves in the effective potential

$$V^2(x) = \left(1 - \frac{1}{x}\right)\left(1 + \frac{\mathscr{L}^2}{x^2}\right) \tag{8}$$

[22]. An ion escapes if $\mathscr{E} \geq 1$ and $\mathscr{E}^2$ exceeds V^2 at all radii beyond its launch position, and it plunges if $\mathscr{E}^2$ exceeds V^2 at all radii between 3 and x along its inward trajectory; otherwise it remains bound to the flow. In the Newtonian case, equations (7)–(8) are replaced by $E = v^2/2 - 1/(2x)$ and $V = -1/(2x) + \mathscr{L}^2/(2x^2)$. In the Monte Carlo integrations, ions that reach $10^3\,r_s$ are classified according to their energy at that radius, and those that remain within it at $t = 2\times10^4\,r_s/c$ are classified at that time.

D. Fluence and yields

Each fluid element accumulates fluence in two stages: during its inflow from $10^3\,r_s$ to r_0 and, if it escapes, along its outbound trajectory. The inflow contribution is obtained by integrating equation (4) along $dt = dr/v_r$; the outbound contribution is accumulated as $\Phi_>(r)\,\Delta t$ at each IAS15 step through a REBOUND heartbeat function. Because the outbound contribution amounts to only $\approx 3\%$ of the total, the yields are determined by the inflow.

The light-element abundances are evolved along the inflow, including both production and proton-induced destruction:

$$\begin{aligned}
d\left(\frac{\rm Li}{\rm H}\right) &= \frac{n_\alpha}{n_{\rm H}}\,\sigma_{\rm Li}\,dF_\alpha + \frac{n_{\rm CNO}}{n_{\rm H}}\,\sigma_{\rm Li}^{\rm CNO}\,dF_p - \sigma_d\,\frac{\rm Li}{\rm H}\,dF_p,\\
d\left(\frac{\rm Be}{\rm O_0}\right) &= \sigma_{\rm Be}\,\frac{\rm O}{\rm O_0}\,dF_p - \sigma_d\,\frac{\rm Be}{\rm O_0}\,dF_p,\\
d\left(\frac{\rm O}{\rm O_0}\right) &= -\sigma_d\,\frac{\rm O}{\rm O_0}\,dF_p,
\end{aligned} \tag{9}$$

which reduces to equation (1) for $\sigma_d F_p \ll 1$. The adopted cross sections, listed in Table I, are representative of the values near their respective peaks [1, 23]; their influence on our results is quantified in Section V. The two lithium channels contribute comparably: although helium is far more abundant than CNO nuclei, the higher reaction threshold places the relevant α particles further into the tail of the Maxwellian. Omitting the CNO channel would lower all lithium predictions by 0.3 dex.

E. Delivery to the companion

The donors in black hole X-ray transients fill their Roche lobes, and each therefore subtends an angle $\theta = R_L/a$ as seen from the black hole (Figure 1a), where R_L/a is given by the approximation of Eggleton [25]. Over an orbit, the companion

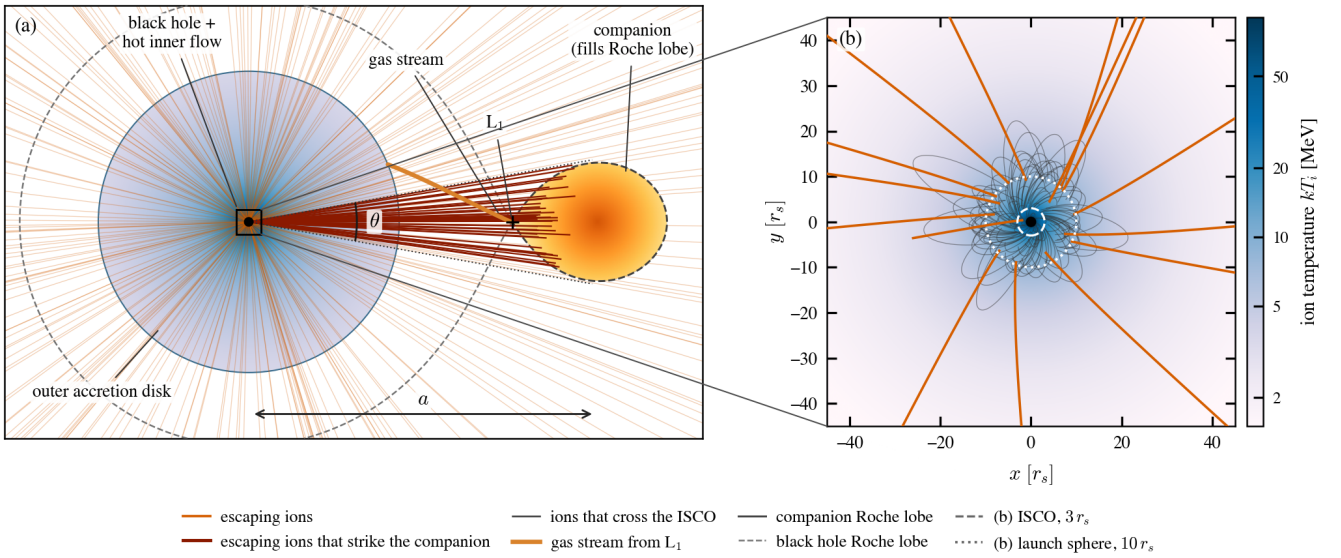


FIG. 1: Geometry of the problem, drawn to scale for A0620−00 ($q = M_2/M_{\rm BH} = 0.061$) in the frame co-rotating with the binary. *(a)* The K-dwarf companion fills its Roche lobe (solid contour through the inner Lagrange point L_1; the dashed contour marks the Roche lobe of the black hole) and feeds the accretion disk, truncated at $0.43\,a$, through a ballistic stream from L_1. Thin lines show the projected directions of the 3230 ions that escape the hot flow from 2×10^5 launches at $10\,r_s$ (exact Schwarzschild classification); the 42 ions that leave within the angle $\theta = R_L/a = 0.18$ of the companion strike it (dark red). *(b)* The hot flow surrounding the black hole: SERPENS trajectories of ions evaporated at $10\,r_s$ (cf. Figure 2), shown over the ion temperature of equation (2). The box in (a) is not to scale; panel (b) spans $3\times10^{-4}\,a$. SERPENS integrates the region shown in (b), and the fraction delivered to the companion follows from the geometry in (a) through equation (10).

TABLE I: Fiducial hot-flow and spallation parameters.

Quantity	Value	Source
Ion temperature kT_i	$0.55\times m_p c^2/(6x)$	[5]
Viscosity, c_1	$\alpha = 0.1$, $c_1 = 0.5$	[5]
Rotation Ω/Ω_K	0.4	[5]
He/H, O/H (number)	0.085, 4.9×10^{-4}	[24]
$\sigma_{\rm Be}$ (p+O, >10 MeV)	1×10^{-26} cm^2	[1]
$\sigma_{\rm Li}$ (α+α, >35 MeV)	5×10^{-26} cm^2	[23]
$\sigma_{\rm Li}^{\rm CNO}$ (p+CNO, >10 MeV)	1×10^{-26} cm^2	[1, 23]
CNO/H (number)	8.3×10^{-4}	[24]
σ_d (destruction by p)	2×10^{-25} cm^2	representative
Li depletion time $\tau_{\rm dep}$	10^7–10^9 yr	[11]

sweeps a band of half-width θ about the orbital plane and occupies a fraction $\theta/4$ of that band at any instant. The fraction of the accreted gas intercepted by the companion is then

$$f_{\rm int} = f_{\rm esc}\, g\, \frac{\theta^2}{4}, \tag{10}$$

where $f_{\rm esc}$ is the escape fraction derived in Section IV A and g is the density of escaping ions near the orbital plane relative to that of an isotropic outflow, measured from the launch directions of the escaping ions. Gravitational focusing by the companion is negligible for ions moving at $\gtrsim 0.1c$.

If the delivered lithium is mixed into a surface convection zone of mass $M_{\rm conv}$ and destroyed on a timescale $\tau_{\rm dep}$, the steady-state surface abundance is

$$\left(\frac{\rm Li}{\rm H}\right)_\star = \frac{\dot M_{\rm Edd}\,\langle \dot m\, y_{\rm Li}(\dot m)\rangle\, f_{\rm int}\,\tau_{\rm dep}}{M_{\rm conv}}, \tag{11}$$

where $A({\rm Li}) = 12 + \log_{10}({\rm Li/H})_\star$ and $y_{\rm Li}(\dot m)$ is the Li/H ratio of the processed material. Since $y_{\rm Li} \propto \dot m$, the delivered lithium scales as $\dot m^2$; after the dynamics, the accretion history is therefore the most important input.

F. Accretion rates and outburst cycles

For A0620−00 we adopt the ADAF fit of Narayan, McClintock & Yi [28], $\dot m = 4\times10^{-4}$. For V404 Cyg, Narayan, Barret & McClintock [32] obtained $\dot M = 1.22\times10^{-9}\,M_\odot$ yr^{-1} for $M = 12\,M_\odot$ at 3.5 kpc; rescaling to $M = 9\,M_\odot$ [30] and to the parallax distance of 2.39 kpc [34] gives $\dot m = 5.0\times10^{-3}$. No ADAF fit has been published for XTE J1118+480. We therefore calibrate the radiative efficiency against the A0620−00 fit at $L/L_{\rm Edd} = 8.5\times10^{-9}$ [35] and the hard-to-soft transition near 2% of the Eddington luminosity [36]:

$$\frac{L}{L_{\rm Edd}} = 0.02\left(\frac{\dot m}{0.02}\right)^P, \qquad P = 3.75. \tag{12}$$

Applied to V404 Cyg, equation (12) yields $\dot m = 1.1\times10^{-3}$, a factor of 4.6 below the direct fit, which indicates the accuracy of this calibration. For XTE J1118+480, with $L_X \approx$

TABLE II: Adopted parameters and observed lithium abundances. $\theta = R_L/a$ for the Roche-lobe-filling donors and R_2/a for the detached Gaia systems; for Gaia BH3, a is the relative separation, $a_1(M_{\rm BH}+M_2)/M_{\rm BH}$, computed from the published $a_1 = 16.17$ AU. The donor mass of XTE J1118+480 follows from $q = 0.027$ [26], and its lithium abundance is an upper limit.

System	$M_{\rm BH}$ [$M_\odot$]	M_2 [$M_\odot$]	$P_{\rm orb}$	θ	$\dot m_q$	outbursts	$M_{\rm conv}$ [$M_\odot$]	observed A(Li)
A0620−00	6.6 [27]	0.40 [27]	7.75 h	0.178	4.0×10^{-4} [28]	1917, 1975	0.15	2.31 ± 0.21 [10]
XTE J1118+480	7.5 [29]	0.20 [26]	4.08 h	0.139	3.3×10^{-4} (eq. 12)	2000, 2005 (hard)	0.20	≤ 1.78 [26]
V404 Cyg	9.0 [30]	0.54 [31]	6.47 d	0.178	5.0×10^{-3} [32]	∼26 yr recurrence	0.30	2.70 ± 0.40 [9]
Gaia BH1	9.62 [12]	0.93 [12]	186 d	3.3×10^{-3}	7×10^{-12} (BHL)	none	0.03	2.3 ± 0.1, normal [12]
Gaia BH2	8.94 [13]	1.07 [13]	3.50 yr	7.3×10^{-3}	2×10^{-9} [13]	none	0.8	not detected (giant) [13]
Gaia BH3	32.7 [14]	0.76 [14]	11.6 yr	1.4×10^{-3}	10^{-8} [33]	none	0.5	no peculiarity [15]

$4\times10^{-9}L_{\rm Edd}$ [37], it yields $\dot m = 3.3\times10^{-4}$. A radiative efficiency that scales linearly with $\dot m$, as is sometimes assumed, would underestimate the accretion rates of all three systems by factors of 40–85.

The transients also spend part of their duty cycle at higher accretion rates during outbursts. We include the hard-state phases of each outburst, below $\approx 2\%$ of the Eddington luminosity where the hot flow is expected to persist [36], assuming an exponential luminosity decline $L = L_0\,e^{-t/\tau_e}$ and computing $\dot m(t)$ from equation (12):

$$\langle \dot m\, y\rangle = \dot m_q\, y(\dot m_q) + \frac{1}{T_{\rm rec}} \sum_{\rm segments} \int \dot m(t)\, y(\dot m(t))\, dt. \qquad (13)$$

A0620−00 underwent outbursts in 1917 and 1975 ($T_{\rm rec} = 57.8$ yr), and the 1975 outburst decayed with $\tau_e = 29$ d [38, 39]. V404 Cyg recurs approximately every 26 yr, most recently in 1989 and 2015 [40]; for this system we adopt the typical value $\tau_e = 30$ d [39]. For both systems we include the decline through the hard state and a hard-state rise of 3 d. XTE J1118+480 has exhibited only hard-state outbursts, in 2000 (peak luminosity 1.2×10^{36} erg s^{-1}) and 2005 (2×10^{35} erg s^{-1}) [41]; we represent these as plateaus of 100 d and 5 d followed by exponential decays of 30 d and 5 d, respectively, and assume one of each per 25 yr.

The adopted system parameters are listed in Table II. The convective-zone masses are estimated from the spectral types and are uncertain by at least a factor of two; they enter equation (11) linearly.

G. The dormant Gaia black holes

The three Gaia black holes are detached and accrete only from the winds of their companions. For Gaia BH1 we use the SERPENS stellar-wind mode: a shifted Maxwellian with the mass-loss rate of Wood et al. [42, 43], $\dot M_w \approx 1.3\times10^9$ kg s^{-1}, and a Parker asymptotic velocity of ≈ 830 km s^{-1} [44, 45]. The Bondi–Hoyle–Lyttleton accretion rate [46, 47],

$$\dot M_{\rm BHL} = \pi\, r_{\rm BHL}^2\, \rho_{\rm wind}\, v_{\rm rel}, \qquad r_{\rm BHL} = \frac{2GM_{\rm BH}}{v_{\rm rel}^2}, \qquad (14)$$

then yields $r_{\rm BHL} \approx 4\times10^9$ m and $\dot M_{\rm BHL} \approx 1\times10^5$ kg s^{-1} at the 1.4 AU separation, corresponding to $\dot m \approx 7\times10^{-12}$. This is consistent with the horizon accretion rate of $\dot m \approx 5\times10^{-12}$ estimated by Rodriguez et al. [48]. For the two giants we adopt published estimates: El-Badry et al. [13] reduce the Bondi–Hoyle–Lyttleton rate onto Gaia BH2 to $\dot m \approx 2\times10^{-9}$ at the horizon, and Cappelluti, Pacucci & Hasinger [33] constrain the rate onto Gaia BH3 to 10^{-9}–10^{-7}, of which we adopt 10^{-8}. None of the three systems has been detected in X-rays. The *Chandra* upper limits are 2.8×10^{29} erg s^{-1} for Gaia BH1 (0.5–7 keV) [48]; $F_X < 5.1\times10^{-15}$ erg s^{-1} cm^{-2}, or $L_X \lesssim 8\times10^{29}$ erg s^{-1} at 1.16 kpc, for Gaia BH2 [13]; and 2.1×10^{29} erg s^{-1} (2–10 keV) for Gaia BH3 [33]. Since the lithium yield scales as $\dot m$ and the delivered lithium as $\dot m^2$, an order-of-magnitude change in any of these rates alters the predicted abundance by two dex, which does not affect our conclusions.

IV. RESULTS

A. Fate of the evaporated ions

Figure 2 shows the trajectories of 1200 ions launched from $10\,r_s$. The great majority return to the black hole: thermal velocities ($v_{\rm sc} \approx 0.1c$ at $10\,r_s$) are approximately one third of the escape speed, and the angular momentum imparted by the rotation of the flow is insufficient to prevent the ions from reaching the ISCO. Of the 1200 ions, 1129 cross the ISCO, 15 escape, and the remaining 56 are still on extended bound orbits at the end of the integration. The escaping ions originate in the high-velocity tail of the Maxwellian and depart on nearly radial trajectories.

Figure 3 shows the escape fraction as a function of launch radius. Exact Schwarzschild geodesics and Newtonian dynamics yield the same escape fraction at all radii, 1.5–1.7%. This agreement is expected: for radial motion, the relativistic escape condition $\mathscr{E} \geq 1$ reduces to $v^2 \geq r_s/r = 2GM/(rc^2)$, which is identical to the Newtonian condition when v is the speed measured by a static observer. The SERPENS Newtonian integrations reproduce the analytic classification on a particle-by-particle basis: in a test with 2000 ions at $10\,r_s$, 99.8% are assigned the same outcome. The escape fraction is nearly independent of r_0 because both kT_i and $v_{\rm esc}^2$ scale as $1/r$, which renders the problem scale-free.

The REBOUNDx 1PN treatment, by contrast, overestimates the escape fraction. At $10\,r_s$ it yields 6.3%, a factor

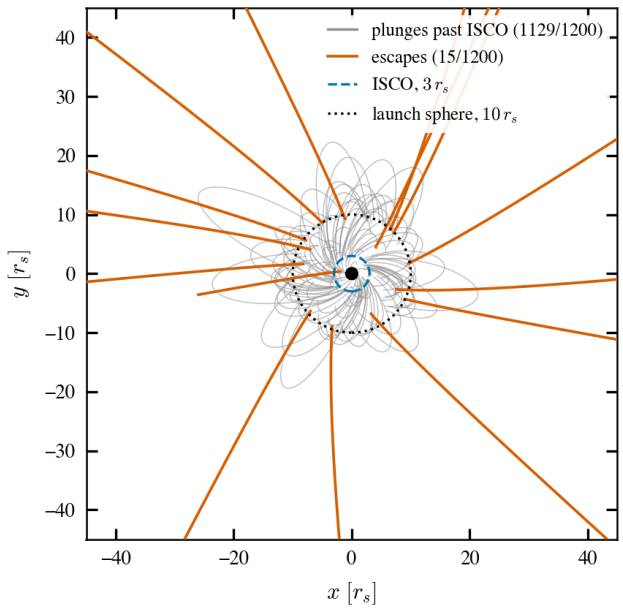


FIG. 2: SERPENS trajectories of 1200 ions evaporated from the hot accretion flow at $r_0 = 10\,r_s$ around a non-spinning black hole, projected onto the orbital plane. Grey curves show a random subset of 120 of the 1129 ions that cross the ISCO (dashed blue circle) and are removed as accreted; red curves show the 15 ions that escape. The black disk marks the horizon and the dotted circle the launch sphere. The integration uses Newtonian gravity and the IAS15 integrator.

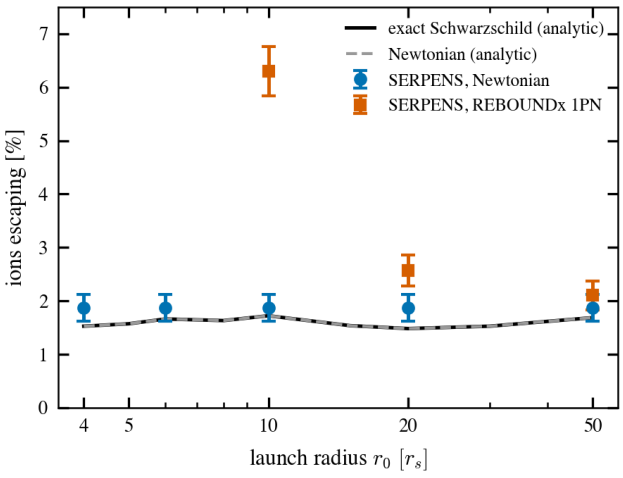


FIG. 3: Percentage of evaporated ions that escape as a function of launch radius. Lines show the analytic classification of 20 000 launches with exact Schwarzschild geodesics (solid) and Newtonian dynamics (dashed), which coincide. Points show SERPENS Monte Carlo integrations of 3000 ions with Newtonian gravity (circles) and with the REBOUNDx 1PN force of equation (6) (squares); error bars are Poisson uncertainties. The 1PN integrations do not converge inside $10\,r_s$.

of 3.7 above the exact value; the discrepancy decreases to factors of 1.7 at $20\,r_s$ and 1.2 at $50\,r_s$. Inside $10\,r_s$ the iterative 1PN solver fails to converge. The origin of the discrepancy is apparent from equation (6): for an ion moving radially outward, the bracketed term becomes $4GM/r + 3v^2$ and is directed outward, so that at $10\,r_s$ and $v = 0.35c$ it cancels 57% of the Newtonian acceleration. The 1PN expansion assumes $v^2/c^2 \ll 1$, a condition that escaping ions violate by construction. Although the 1PN treatment remains accurate at binary separations, where $v/c \sim 10^{-3}$ (Appendix A), it should not be applied to ballistic motion within $\sim 20\,r_s$. In what follows we adopt the exact Schwarzschild classification, with $f_{\rm esc} = 1.55\%$ and an equatorial enhancement $g = 1.22$ arising from the rotation of the flow.

B. Spallation yields

Figure 4 shows the yields of material reaching r_0 as a function of the accretion rate. Both yields scale linearly with $\dot{m}$ until destruction becomes significant near $\dot{m} \sim 10^{-2}$. A fluid element requires approximately 40 s to fall from 10^3 to $10\,r_s$ around a $7\,M_\odot$ black hole. At the quiescent accretion rate of A0620−00, this exposure corresponds to a fluence of 1.4×10^{23} cm^{-2} above 10 MeV and a Be/O ratio of 1.4×10^{-3}. For comparison, the ratio of 10^{-5} inferred by Doyle et al. for GALEX J2339−0424 requires $\sim 10^7$ yr of exposure in a Jovian radiation belt, whereas the hot flow attains it within a single inflow time for $\dot{m} \gtrsim 3 \times 10^{-6}$.

C. Comparison with the observed lithium

Equation (10) gives $f_{\rm int} = 1.5 \times 10^{-4}$ for A0620−00 and V404 Cyg, whose mass ratios are nearly identical, and 0.9×10^{-4} for XTE J1118+480, whose donor is smaller. These values are comparable to the upper end of the range 10^{-6}–10^{-4} required by Yi & Narayan [11]. In their notation the escaping fraction was $F_{\rm esc} \approx 10^{-3}\,(10^8\,{\rm yr}/\tau_{\rm dep})$; our value of $f_{\rm esc} = 1.5 \times 10^{-2}$ corresponds to the short end of their range of depletion times. The predictions are compared with the measurements in Figure 5 and Table III.

When outbursts are included, the predicted ranges encompass the observed lithium abundances of A0620−00 and V404 Cyg, and the computed $f_{\rm int}$ lies within the required range for both systems. For XTE J1118+480 the measurement is an upper limit, $A({\rm Li}) \le 1.78$, which the model satisfies for $\tau_{\rm dep} \lesssim 10^8$ yr. The implied depletion times differ between the systems: $\tau_{\rm dep} \approx 4 \times 10^7$ yr for the subgiant in V404 Cyg and $\approx 3 \times 10^8$ yr for the K dwarf in A0620−00. This ordering is consistent with stellar structure; Yi & Narayan [11] quote depletion times of a few $\times 10^8$ yr for main-sequence K dwarfs and $\sim 5 \times 10^7$ yr for giants. The importance of the outbursts also differs between the systems. In V404 Cyg the quiescent accretion rate is already $\dot{m} = 5 \times 10^{-3}$, and the outbursts increase the prediction by only 0.03 dex. In A0620−00 the quiescent flow alone falls 0.2 dex short even for $\tau_{\rm dep} = 10^9$ yr, and the hard-state phases of the outbursts raise the prediction by 0.7 dex. We emphasise that no parameter was adjusted to

TABLE III: Predicted and observed lithium abundances of the companions, and predicted beryllium abundances, for $r_0 = 10\,r_s$. Ranges span $\tau_{\rm dep} = 10^7$–10^9 yr. $f_{\rm req}$ is the delivered fraction required to reproduce the observed lithium with outbursts included (an upper limit for XTE J1118+480). The final column gives the delivered beryllium, expressed as A(Be), when the model reproduces the observed lithium; the solar value is 1.38 [24].

System	$f_{\rm int}$ (SERPENS)	$f_{\rm req}$	A(Li), quiescent	A(Li), + outbursts	observed A(Li)	predicted A(Be)
A0620−00	1.5×10^{-4}	4×10^{-5}–4×10^{-3}	0.1 to 2.1	0.9 to 2.9	2.31 ± 0.21	1.6
XTE J1118+480	0.9×10^{-4}	$\leq 9\times10^{-6}$–9×10^{-4}	−0.3 to 1.7	0.8 to 2.8	≤ 1.78	≤ 1.1
V404 Cyg	1.5×10^{-4}	6×10^{-6}–6×10^{-4}	2.1 to 4.1	2.1 to 4.1	2.70 ± 0.40	2.1
Gaia BH1	5×10^{-8}	–	−18 to −16	–	2.3 ± 0.1 (intrinsic)	–
Gaia BH2	2.5×10^{-7}	–	−14 to −12	–	not detected	–
Gaia BH3	9×10^{-9}	–	−13 to −11	–	normal	–

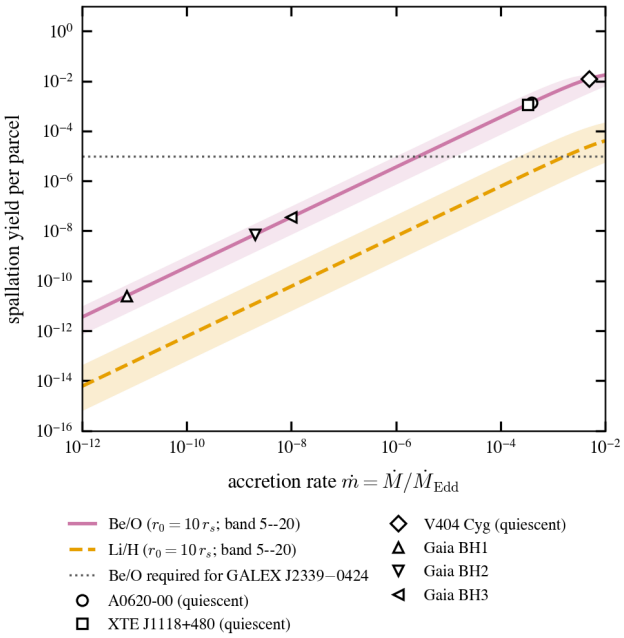


FIG. 4: Spallation yields of processed material as a function of the Eddington-scaled accretion rate: beryllium per initial oxygen nucleus (solid) and lithium per hydrogen nucleus (dashed) for $r_0 = 10\,r_s$, with shaded bands spanning $r_0 = 5$–$20\,r_s$. The dotted line marks the Be/O ratio inferred by Doyle et al. [1] for the polluted white dwarf GALEX J2339−0424. Markers indicate the quiescent accretion rates adopted for the six black holes.

obtain this agreement: all inputs listed in Tables I and II were fixed prior to the comparison.

D. The dormant Gaia black holes

The Gaia black holes represent the opposite limit. Their accretion rates are four to nine orders of magnitude below those of the transients, and the yields per fluid element are correspondingly lower (Figure 4). Their companions are also more distant, subtending $\theta = 3.3\times10^{-3}$ (Gaia BH1), 7.3×10^{-3} (the larger giant in Gaia BH2) and 1.4×10^{-3} (Gaia BH3), so that $f_{\rm int}$ ranges from 10^{-8} to 3×10^{-7} (Figure 5b). The delivered lithium corresponds to $A(\mathrm{Li}) \approx -17$ for Gaia BH1 and -14 to -11 for the two giants (Table III), at least ten orders of magnitude below the detection threshold.

These predictions are consistent with the observations. The companion of Gaia BH1 does contain lithium, $A(\mathrm{Li}) = 2.3\pm0.1$, but El-Badry et al. [12] show that this abundance is typical of a solar-type star older than the Hyades, and our model attributes none of it to the black hole. The giant in Gaia BH2 shows no lithium, as expected after the first dredge-up [13], and the giant in Gaia BH3 shows no peculiarity in 29 elements from lithium to thorium [15]. The companion of Gaia BH2 is nevertheless α-enhanced, $[\alpha/\mathrm{Fe}] = +0.26\pm0.05$, despite its thin-disk kinematics [13]. This enhancement cannot arise from spallation, which produces Li, Be and B and depletes rather than enriches oxygen. El-Badry et al. estimate that direct capture of supernova ejecta at this separation would deliver at most $7\times10^{-6}\,M_\odot$ of oxygen, which is insufficient, and suggest that slow, bound ejecta might contribute if modelled in detail. The α-enhancement and the lithium abundance therefore trace distinct processes, and a lithium or beryllium excess in the companion of a Gaia black hole would require an origin other than the mechanism considered here.

E. A beryllium test

Spallation of oxygen in the same flow also produces beryllium. In the linear regime, the Be/Li ratio of the processed material is set by the ratio of the proton and α fluences, and the delivered beryllium abundance is 0.6–0.7 dex below that of lithium (Table III). If the model reproduces the observed lithium abundances, the beryllium delivered to the companion of V404 Cyg corresponds to $A(\mathrm{Be}) \approx 2.1$, five times the solar value. For A0620−00 the delivered beryllium, $A(\mathrm{Be}) \approx 1.6$, is comparable to the initial abundance and would raise the surface abundance by a factor of about two, whereas for XTE J1118+480 the delivered abundance falls below the solar value and would not be detectable. Because beryllium is destroyed at higher temperatures than lithium, its depletion time is longer, and these estimates are lower limits within the model. The Be II resonance doublet at 3130–3131 Å is accessible from the ground for the brighter companions and provides the direct counterpart to the beryllium measurement that motivated the analysis of polluted white dwarfs.

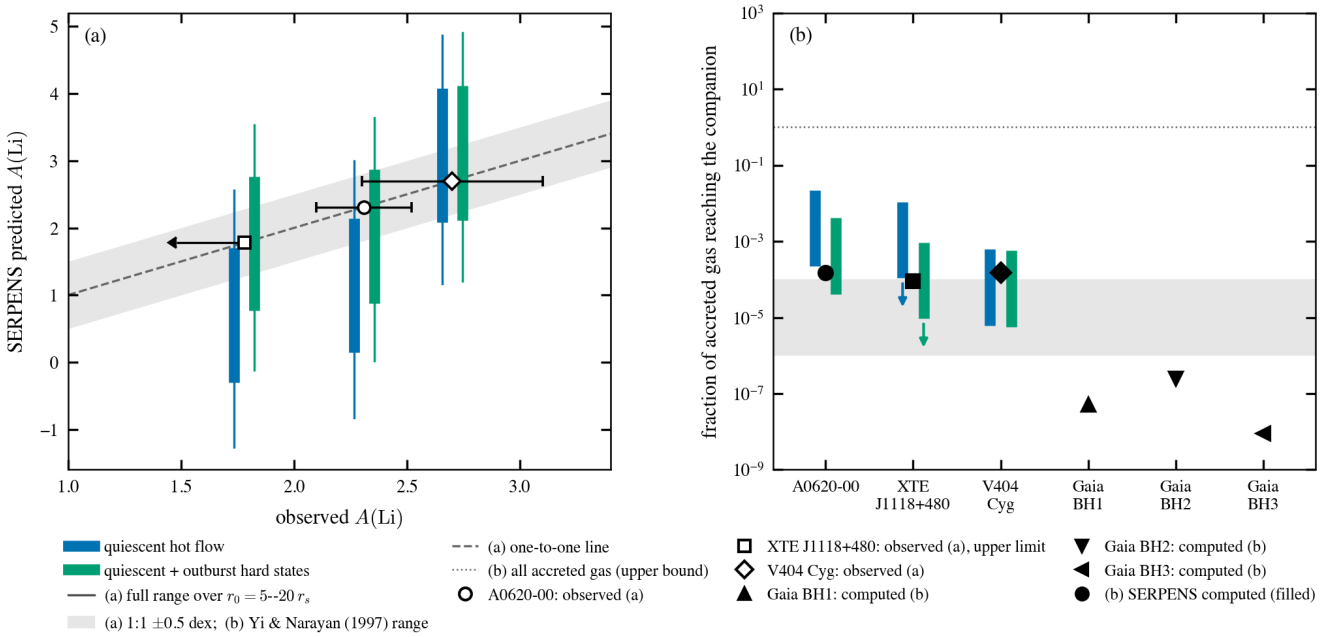


FIG. 5: Comparison of the SERPENS predictions with observations. *(a)* Predicted surface lithium abundance of the companion against the measured value. Thick bars span depletion times $\tau_{\rm dep} = 10^7$–10^9 yr at $r_0 = 10\,r_s$, and thin lines extend over $r_0 = 5$–$20\,r_s$. Blue bars correspond to the quiescent hot flow alone and green bars include the hard-state phases of the observed outbursts. Open symbols mark the measurements on the one-to-one line, the arrow denotes the upper limit for XTE J1118+480, and the grey band spans ±0.5 dex. *(b)* Fraction of the accreted gas that must reach the companion to reproduce the measured lithium (bars, with the same colours and range of $\tau_{\rm dep}$) and the fraction computed by SERPENS from equation (10) (filled symbols). The grey band indicates the range required by Yi & Narayan [11], and the dotted line corresponds to delivery of all accreted gas. Arrows mark the requirement for XTE J1118+480, which is an upper limit. The Gaia black holes are shown with their computed fractions only, as none of their companions shows a lithium excess.

V. DISCUSSION

A. Interpretation of the agreement

The model comprises three components. The dynamics determine $f_{\rm esc}$ and g; this component has been verified against exact geodesics and is reliable to within the Monte Carlo uncertainty of a few percent. The geometry determines θ, which is fixed by the measured mass ratios. The principal uncertainties lie in the microphysics of the hot flow and in the response of the companion. Table IV lists the change in the fiducial prediction ($r_0 = 10\,r_s$, outbursts included, $\tau_{\rm dep} = 10^8$ yr) when individual inputs are varied.

The ion temperature is the largest single source of uncertainty. Because the lithium-producing α particles lie in the exponential tail above 35 MeV, increasing T_i from 55% to 100% of the virial value raises the prediction by 1.5 dex. Hot accretion flows are also expected to lose mass through winds [6], which would increase $f_{\rm esc}$ above our single-attempt ballistic estimate. Both effects increase the predicted abundance; given the agreement of the fiducial model with the data, either would require shorter depletion times than those inferred above. The agreement shown in Figure 5 should therefore be regarded as an order-of-magnitude result: the mechanism delivers the observed quantity of lithium for plausible parameters without tuning, but the comparison does not constrain any individual parameter.

TABLE IV: Change in the predicted A(Li) when individual inputs are varied, for $r_0 = 10\,r_s$ with outbursts included and $\tau_{\rm dep} = 10^8$ yr.

Change	ΔA(Li) [dex]
kT_i: 0.55 → 1.0 of virial	+1.5
$\sigma_{\rm Li}$ (α+α) ×2	+0.2
$\tau_{\rm dep}$: $10^8 \to 10^9$ yr	+1.0
$f_{\rm esc}$ × 10 (MHD winds)	+1.0
$M_{\rm conv}$ × 2	−0.3
r_0: $10 \to 5\,r_s$	+0.8
r_0: $10 \to 20\,r_s$	−0.9

B. Limitations

The ions in our model are ballistic after launch. In an ADAF, the magnetic field couples the ions to the flow, and material escapes through winds and jets whose launching we do not model; the escape fraction of 1.5% should therefore be interpreted as the fraction of processed material available to an outflow. We further assume that the companion intercepts the outflow geometrically, neglect the companion's own wind

and magnetosphere, and represent the outbursts by exponential light curves rather than the observed profiles. The lithium depletion time is the least constrained quantity in the problem and spans two orders of magnitude. Finally, we do not include lithium production during the soft states, which Yi & Narayan [11] suggested may contribute; its inclusion would strengthen our conclusions.

C. Alternative explanations

Two alternative origins have been proposed for the lithium. The lithium could have been synthesised in the supernova that formed the black hole and captured by the companion, as has been proposed for the α-element excesses in some of these systems [10, 26]. A K or M dwarf with a deep convection zone would, however, have destroyed this lithium over the $\gtrsim 10^8$ yr since the explosion. Alternatively, the systems could be sufficiently young to have retained their primordial lithium; González Hernández et al. [10] note that this would require A0620−00 to be as young as the Pleiades. Continuous production in the hot flow avoids both difficulties because the lithium is continually replenished. The beryllium prediction distinguishes our model from the primordial explanation, since a young star would exhibit a solar beryllium abundance rather than the fivefold excess predicted for V404 Cyg.

D. The lithium isotope ratio

The $^6\mathrm{Li}/^7\mathrm{Li}$ ratio has been proposed as a means of distinguishing ongoing production from preserved lithium. Casares et al. [9] measured $^6\mathrm{Li}/^7\mathrm{Li} = 0.12^{+0.08}_{-0.05}$ in the neutron star binary Cen X-4 and interpreted the low ^{6}Li abundance as favouring preservation, while noting that production could not be excluded. Two considerations limit the diagnostic power of this ratio for our model. First, the production ratio depends on the ion temperature: $\alpha+\alpha$ reactions in a hot flow yield $0.1 < {}^6\mathrm{Li}/^7\mathrm{Li} < 10$ for kT_i between 3 and 50 MeV, and the value measured in Cen X-4 corresponds to $kT_i \approx 4$ MeV [9], within the 4–20 MeV range of our flow. Second, ^{6}Li is destroyed at lower temperatures than ^{7}Li in stellar interiors [49]. In steady state, the abundance of each isotope equals its delivery rate multiplied by its depletion time, so that the surface ratio equals the production ratio multiplied by $\tau_{\mathrm{dep}}(^6\mathrm{Li})/\tau_{\mathrm{dep}}(^7\mathrm{Li}) < 1$. Freshly delivered lithium is therefore also expected to appear ^{6}Li-poor at the surface. Because we do not model the isotopic cross sections or the burning within the companion in sufficient detail to predict the ratio, we regard it as non-discriminating; beryllium, as discussed above, provides a more decisive test.

Casares et al. also note that lithium has been detected only in companions of spectral type earlier than K5, and never above the meteoritic value $A(\mathrm{Li}) = 3.3$ [9]. Both observations are consistent with our model as well as with preservation. Later spectral types possess deeper convection zones and shorter depletion times, which reduce the delivered lithium through equation (11), and the depletion times we infer, $\lesssim 3\times10^8$ yr, keep the steady-state abundance below the meteoritic value.

E. Future work

The most informative next step is observational: near-ultraviolet spectroscopy of the companion of V404 Cyg to measure its beryllium abundance. On the theoretical side, the escape fraction should be recomputed with an MHD wind launched from the hot flow, and the response of the companion should include rotational mixing, which governs τ_{dep}. Black hole spin moves the ISCO inward and raises the ion temperatures in its vicinity; replacing equation (8) with Kerr geodesics would allow a test of whether the lithium enhancement correlates with spin. Finally, Gaia DR4 is expected to reveal many additional dormant black holes [12], whose companions our model predicts will show no spallation lithium.

VI. CONCLUSIONS

We have extended the spallation argument of Doyle et al. [1] for polluted white dwarfs to black hole binaries, using SERPENS to follow ions evaporated from the hot accretion flow. Our principal conclusions are as follows.

1. Approximately 1.5% of the evaporated ions escape, while the remainder cross the ISCO. Exact Schwarzschild geodesics and Newtonian dynamics yield the same escape fraction, whereas the REBOUNDx 1PN approximation overestimates it by a factor of 3.7 at $10\,r_s$ and fails to converge at smaller radii.

2. Gas falling through the hot flow of a quiescent black hole accumulates, within ≈ 40 s, the proton fluence that Doyle et al. attribute to $\sim 10^7$ yr of exposure in a Jovian radiation belt.

3. The fraction of accreted gas delivered to the companion is $(0.9\text{–}1.5)\times10^{-4}$ for A0620−00, XTE J1118+480 and V404 Cyg, comparable to the upper end of the range required by Yi & Narayan, who treated this quantity as a free parameter.

4. Using published ADAF accretion rates and the observed outburst histories, the predicted surface lithium abundances agree with the measurements for A0620−00 and V404 Cyg and with the upper limit for XTE J1118+480, for depletion times of 4×10^7 yr (subgiant) to 3×10^8 yr (K dwarf).

5. The dormant Gaia black holes deliver at least ten orders of magnitude too little lithium to be detected, in agreement with the observed companions: the lithium in the Gaia BH1 G dwarf is consistent with its age, the Gaia BH2 giant shows none, and the Gaia BH3 giant exhibits no peculiarities.

6. The model predicts a beryllium abundance approximately five times the solar value in the companion of V404 Cyg, which can be tested with near-ultraviolet spectroscopy. The lithium isotope ratio does not discriminate between production and preservation, because ^{6}Li is destroyed before ^{7}Li, whereas the beryllium abundance does.

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Appendix A: Integrator benchmark at binary scales

Prior to the strong-field calculations, we tested the GR-extended SERPENS integrator on a benchmark system, `StellarBH-10`, consisting of a $10\,M_\odot$ black hole and a mass-losing source on a near-circular orbit at $a = 5\times10^8$ m ($1.7\times10^4\,r_s$). Hydrogen is sputtered from the source following the Smyth–Combi distribution [50], with bulk and maximum velocities of 5% and 30% of the local orbital velocity. We integrated eight orbits of the source using the Wisdom–Holman scheme [51] together with the REBOUNDx `gr_full` force [20, 52]. At these separations $v/c \sim 10^{-3}$, and the 1PN expansion is accurate. The particles form a ring along the source orbit (Figure 6), and none reach the ISCO. Over two orbits of the Gaia BH1 configuration, the relative energy error is $|\Delta E/E| \simeq 3.6\times10^{-9}$. These tests demonstrate that the integrator is reliable at binary scales; Section IV A identifies where the 1PN treatment ceases to be adequate.

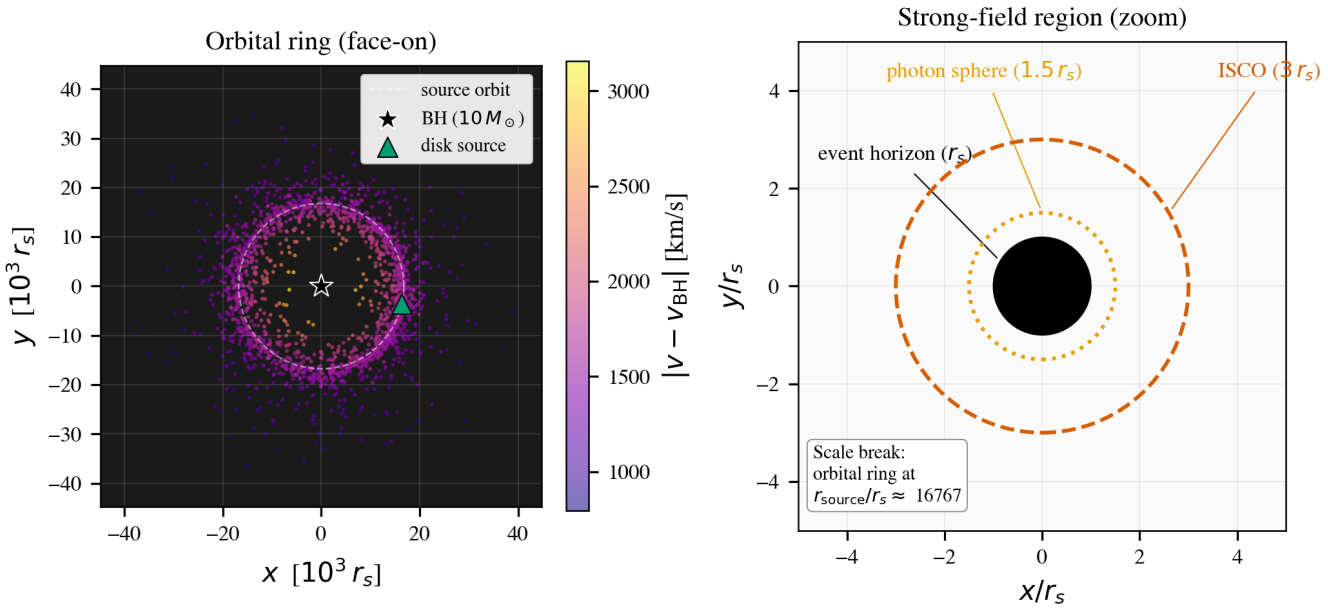


FIG. 6: Particle distribution in the orbital plane of the `StellarBH-10` benchmark. *Left:* full view showing the source (green triangle) and the ring of ~ 1900 test particles, coloured by speed relative to the black hole. *Right:* the strong-field region, showing the horizon at r_s (filled), the photon sphere at $1.5\,r_s$ (dotted) and the ISCO at $3\,r_s$ (dashed red) [16]. Distances are in units of r_s.